

# Software Requirements Specification (SRS)

## Project Title:

**JEEVAN SETU – Intelligent Patient Transfer Decision Support System  
From ICU to HDU & HDU to ICU**

## 1. Introduction

### 1.1 Purpose

The purpose of this system is to assist healthcare professionals in determining whether a patient in the **Intensive Care Unit (ICU)** is stable enough to be transferred to the **High Dependency Unit (HDU)**.

The system analyzes real-time physiological parameters and calculates an **Early Warning Score (EWS)** to evaluate patient stability. Based on this score, the system provides recommendations such as:

- Fit for Transfer (HDU)
- Re-evaluate Patient
- Keep in ICU

This system is designed to **support clinical decision-making**, not replace doctors.

### 1.2 Scope

The proposed system is a **web-based decision support system** that collects patient vital signs, calculates a stability score, and generates transfer recommendations.

The system includes:

- Patient registration and management
- Entry of patient vital signs

- Early Warning Score calculation
- Alert generation for unstable patients
- ICU monitoring dashboard
- Transfer decision support
- Attendant dashboard
- Explainable AI
- Chatbot
- Report generator

The system can be integrated with hospital monitoring devices in the future but currently uses **manual or simulated data input**.

### 1.3 Definitions and Abbreviations

| Term | Meaning              |
|------|----------------------|
| ICU  | Intensive Care Unit  |
| HDU  | High Dependency Unit |
| EWS  | Early Warning Score  |
| RR   | Respiratory Rate     |
| HR   | Heart Rate           |
| BP   | Blood Pressure       |

## 2. Overall Description

### 2.1 System Perspective

The system acts as an **add-on clinical decision support tool** in a hospital environment.

The workflow is:

Patient Monitoring → Data Analysis → Score Calculation → Alert Generation → Transfer Recommendation

The system helps doctors make **consistent and data-driven decisions**.

## **2.2 Product Functions**

The main functions of the system are:

1. User authentication and login
2. Patient registration and record management
3. Entry of patient vital parameters
4. Early Warning Score calculation
5. Alert generation for abnormal vitals
6. ICU monitoring dashboard
7. Transfer recommendation system
8. Patient history and report viewing
9. Attendant dashboard
10. Explainable AI
11. Chatbot
12. Report generator

## **2.3 User Classes**

### **Doctor**

Responsibilities:

- View patient dashboard
- View alerts
- Review transfer recommendations
- Monitor patient condition

## **Nurse**

### **Responsibilities:**

- Enter patient vital signs
- Update patient data
- Monitor alerts

## **Admin**

### **Responsibilities:**

- Manage user accounts
- Manage system configuration
- View overall ICU statistics

## **Attendant (Patient Caregiver / Relative)**

The attendant is a non-medical user associated with the patient.

### **Responsibilities:**

- View patient basic details
- View simplified patient status (Stable / Warning / Critical)
- View transfer recommendation
- Receive alerts and updates

### **Restrictions:**

- Cannot edit data
- Cannot access detailed medical information
- Limited access via role-based permissions

## **2.4 Operating Environment**

The system will run in a hospital IT environment.

### **Platform:**

- Web Browser
- Windows / Linux systems

### Technology Stack:

Frontend

HTML

CSS

Typescript

Bootstrap

Backend

Python

Flask

Database

MySQL

### 2.5 Design Constraints

- The system relies on **manual input of patient vitals**.
- Clinical scoring is based on **predefined thresholds**.
- Real-time monitoring is simulated using periodic updates.

### **3. System Features**

#### **3.1 User Authentication**

##### **Description**

Users must log in to access the system.

##### **Functional Requirements**

The system shall:

- allow users to log in
- authenticate using username and password
- redirect users to role-specific dashboards
- allow users to log out

#### **3.2 Patient Management**

##### **Description**

This module allows registration and management of ICU patients.

##### **Functional Requirements**

The system shall allow users to:

- add new patient records
- view patient details
- update patient information
- display patient list

##### **Patient Information**

Patient ID

Name

Age

Gender

Diagnosis

ICU Bed Number

Admission Date

Doctor Assigned

### **3.3 Vital Sign Entry**

#### **Description**

Nurses can enter patient vital parameters into the system.

#### **Vital Parameters Collected**

Respiratory Rate

Heart Rate

Systolic Blood Pressure

Temperature

Level of Consciousness (AVPU)

The system stores these values in the database.

### **3.4 Early Warning Score Calculation**

The system calculates a patient stability score using the following parameters:

Respiratory Rate

Heart Rate

Systolic Blood Pressure

Temperature

AVPU Consciousness Level

#### **Score Calculation**

Total Score = RR + HR + BP + Temperature

Maximum Score = 14

### 3.5 Score Interpretation

| Score | Patient Condition | Recommendation       |
|-------|-------------------|----------------------|
| 0–2   | Stable            | Fit for HDU Transfer |
| 3–4   | Moderate Risk     | Re-evaluate          |
| ≥5    | Critical          | Keep in ICU          |

### 3.6 Alert System

The system generates alerts when patient vitals exceed safe thresholds.

#### Alert Types

Critical Alert

Warning Alert

Stable Status

#### Alert Example

ALERT

Patient unstable

Score = 6

Recommendation: Keep in ICU

Alerts are displayed on the dashboard.

### 3.7 ICU Monitoring Dashboard

The dashboard provides a real-time overview of ICU patients.

#### Dashboard Information

Total ICU Patients

Stable Patients

Warning Patients

Critical Patients

## **Patient Table**

Patient ID

Vital Parameters

Score

Status

Status Colors

Green – Stable

Yellow – Warning

Red – Critical

## **3.8 Transfer Recommendation System**

Based on the Early Warning Score, the system recommends:

Fit for Transfer

Re-evaluate Patient

Keep in ICU

The recommendation includes reasons based on abnormal parameters.

## **3.9 Patient History**

The system stores all patient vital records and allows doctors to view:

- historical vitals
- previous scores
- past alerts

## **3.11 Attendant Dashboard**

Provides simplified information for patient caregivers.

### **Features:**

- Patient basic details
- Status (Green / Yellow / Red)
- Transfer recommendation
- Alerts in simple language
- Last updated time

### **3.12 Explainable AI**

Explains decision using parameter deviations.

### **3.13 Chatbot**

Provides quick Q&A.

### **3.14 Report Generator**

Generates patient reports automatically.

## **4. Workflow of the System**

The workflow of the system is as follows.

Step 1:User logs into the system.

Step 2:Nurse registers patient in ICU.

Step 3:Nurse enters patient vital parameters.

Step 4;system calculates Early Warning Score.

Step 5:System checks score against thresholds.

Step 6:System generates alerts if necessary.

Step 7:Dashboard updates patient status.

Step 8:system recommends ICU or HDU transfer.

## **5. System Architecture**

System Architecture consists of three layers:

Presentation Layer

Frontend user interface

Application Layer

Backend logic and decision engine

Data Layer

Database storing patient records

Flow:

User Interface → Backend API → Decision Engine → Database → Dashboard

## 6. Non-Functional Requirements

### Performance

The system should process patient data and generate recommendations within **2 seconds**.

### Security

- User authentication required
- Role-based access control
- Patient data protection

### Reliability

The system should maintain accurate records and avoid data loss.

### Usability

The interface must be simple and easy for healthcare staff to use.

### Scalability

The system should support additional hospital departments in the future.

## 7. Database Requirements

### Users Table

User ID

Name

Email

Password

Role

### **Patients Table**

Patient ID

Name

Age

Gender

Diagnosis

Bed Number

### **Vitals Table**

Record ID

Patient ID

Respiratory Rate

Heart Rate

Blood Pressure

Temperature

Score

Timestamp

### **Alerts Table**

Alert ID

Patient ID

Alert Type

Message

Timestamp

### **Attendant Access Table**

Stores mapping between patient and attendant.

#### **Fields:**

- attendant\_id
- patient\_id
- name
- relation (father/mother/etc.)
- contact\_number
- login\_credentials (optional)

#### **Purpose:**

- Allows attendants to view only their patient
- Maintains privacy and access control

### **AI Explanation Table**

Stores generated explanations for decisions.

#### **Fields:**

- explanation\_id
- patient\_id
- abnormal\_parameters
- explanation\_text
- decision
- timestamp

#### **Example:**

- “High HR, Low BP → Unstable”

#### **Purpose:**

- Supports Explainable AI
- Used in dashboard & reports

### **Chatbot Interaction Table**

Stores chatbot queries and responses.

#### **Fields:**

- chat\_id
- patient\_id
- user\_query
- bot\_response
- timestamp

#### **Purpose:**

- Tracks chatbot usage
- Improves system debugging

Stores generated reports.

**Fields:**

- report\_id
- patient\_id
- vitals\_snapshot
- score
- decision
- explanation
- generated\_time

**Purpose:**

- Stores reports for download
- Maintains history

**Audit Log Table**

Tracks system actions.

**Fields:**

- log\_id
- user\_id
- action
- timestamp

 Useful for:

- Security
- Debugging

**8. Expected Outcomes**

The system will:

- improve ICU resource management
- reduce ICU overcrowding
- provide objective patient evaluation
- support faster clinical decision making
- enhance patient safety

- AI transparency

- Faster decision understanding
- Automated reporting